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Prelab 3.1 IoT Data Collection 3 - Industry data communication protocols: Power meter (Modbus) and vibration (IO-Link)

Learning Goals

Students will be able to:

1. Define and describe Modbus protocol
2. Define and describe IO-Link protocol
3. Explain TCP/IP and REST API

1.1 Introduction

Seamless real-time communication is the core prerequisite for all Industry 4.0 and Industrial Internet of Things (IIoT) implementations. In Lab3, we will collect data from two IoT devices, a power meter (Acuvim L series, AccuEnergy, Canada), and a vibration sensor (VVB001, IFM, Germany) with a IO-Link master (AL1302, IFM, Germany). Please visit and look into both web pages of the devices. For the power meter, Modbus protocol will be used. On the other hand, for the vibration sensor, the IO-Link protocol will be used. When we read data from the vibration sensor through the IO-Link master, we use REST API via TCP/IP (so-called EtherNet/IP) communication. But the communication between the vibration sensor and the IO-Link master is being made by IO-Link protocol. This prelab is to install a related package on Raspberry Pi and to be familiar with the industry standard protocols. Modbus and IO-Link are representative protocols in Fieldbus communications. And TCP/IP is a representative protocol in Industrial Ethernet communications.

On top of that, it is highly recommended to go through JSON data format and how to parse it from online resources.

1.2 Python package installation on Raspberry Pi

'pymodbus' package installation

Install 'mymodbus' package on your Raspberry Pi as below.

Raspberry Pi - Terminal

```
pip3 install pymodbus==2.5.3
```

After installing 'pymodbus' package, see if it is installed successfully using the command 'pip3 list'.

Task 1.1

Capture your terminal window as Figure 1 after running 'pip3 list'.

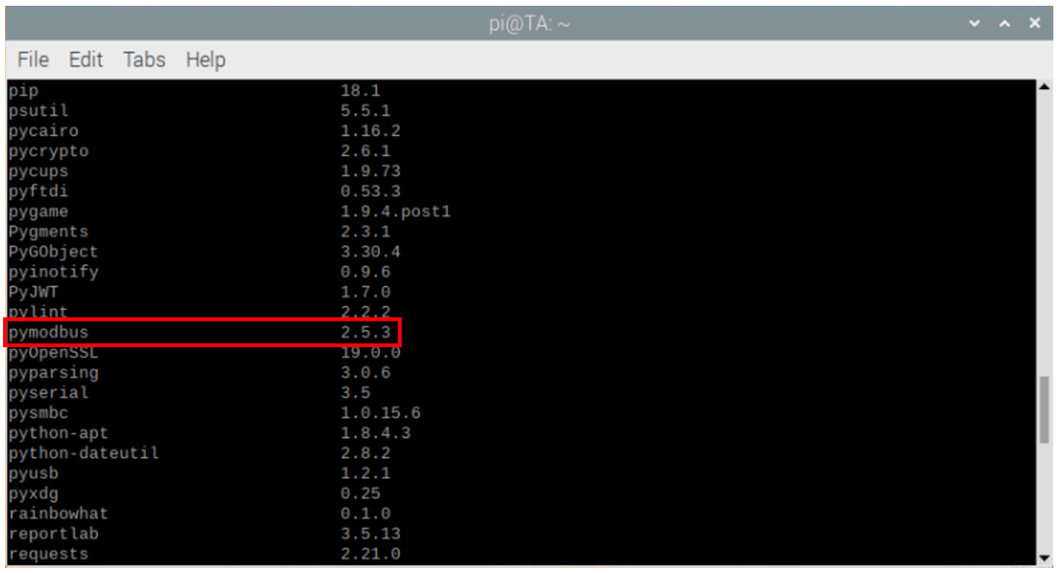
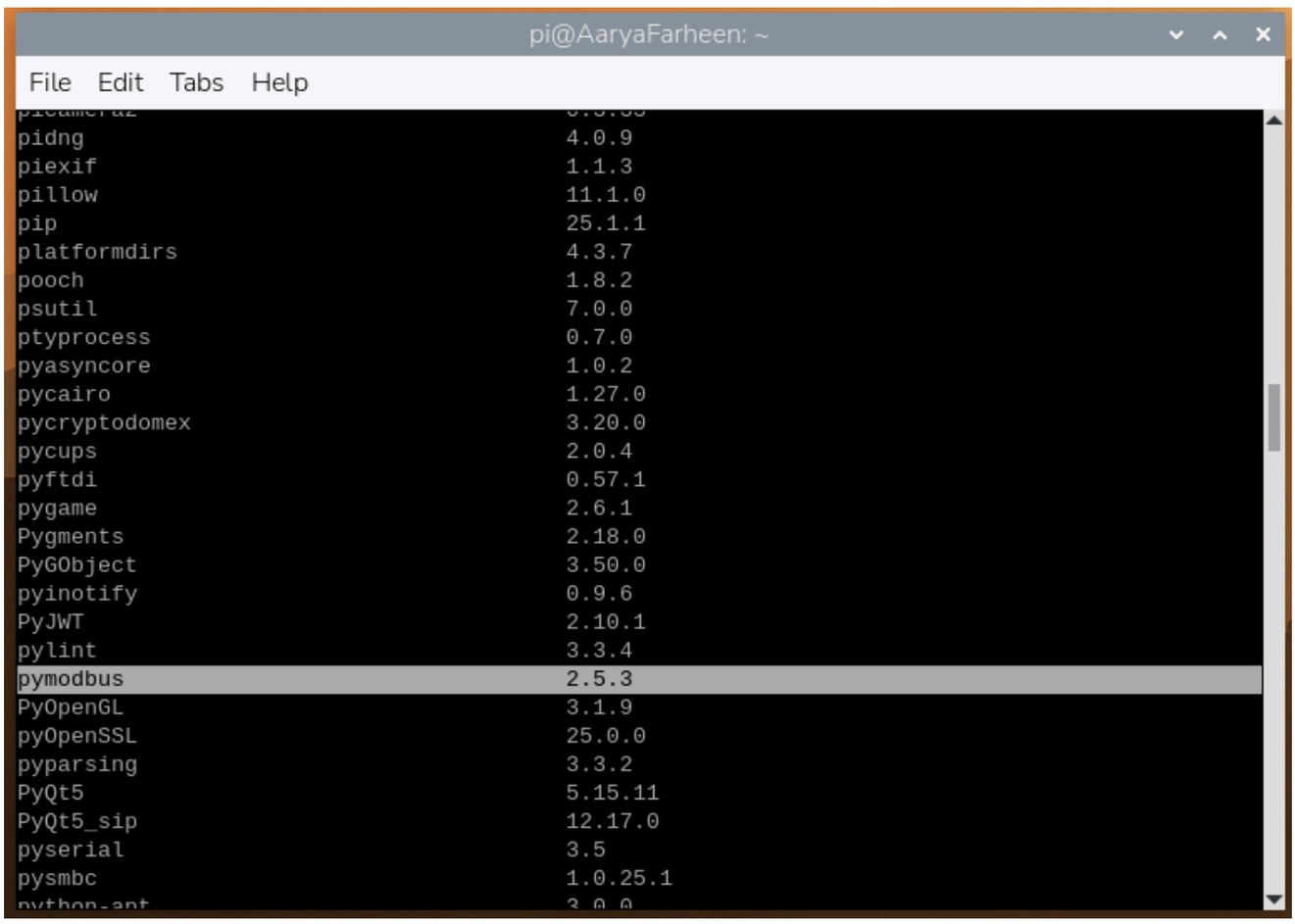


Figure 1 Checking the installed Python packages



Task 1.2

Address the following questions by checking provided resources.

```
from IPython.display import YouTubeVideo
YouTubeVideo('txi2p5_OjKU')

#Watch this video to answer question 1.
```

1) Explain what is Modbus protocol?

Modbus is a communication protocol widely used in industrial automation systems to connect electronic devices such as sensors, controllers, and actuators. It allows devices to exchange data

```
from IPython.display import YouTubeVideo
YouTubeVideo('YDbWt_wE3YE')

#Watch this video to answer question 2.
```

2) Explain what is IO-Link protocol?

IO-Link is a point-to-point communication protocol used in industrial automation to connect sensors and actuators to a control system. Unlike traditional binary signals, IO-Link allows devices to send digital data, such as measurement values, diagnostics, and configuration parameters, over a standard three-wire connection.

```
from IPython.display import YouTubeVideo
YouTubeVideo('614QGgw_FA4')

#Watch this video to answer question 3.
```

3) Explain what is TCP/IP protocol?

TCP/IP (Transmission Control Protocol / Internet Protocol) is a set of networking protocols that allows devices to communicate over the internet or local networks. It defines how data is packaged, addressed, transmitted, and received between computers and devices.

```
from IPython.display import YouTubeVideo
YouTubeVideo('lsMQRaeKNDk')

#Watch this video to answer question 4.
```

4) Explain what is REST API?

REST API (Representational State Transfer Application Programming Interface) is a protocol that allows different software systems to communicate over the internet using standard HTTP methods. It enables clients (like web or mobile applications) to request data or perform actions on a server.

5) What can be the applications of power and vibration monitoring in the industry?

Power and vibration monitoring enables industries to reduce maintenance costs, improve operational efficiency, enhance safety, and maintain product quality by continuously tracking machine performance.

Please continue to Prelab 3.2 here.